

FACULTY OF ENGINEERING AND TECHNOLOGY  
BACHELOR OF TECHNOLOGY

Image Processing Laboratory  
(303105382)

SEMESTER VII  
Department of AI & AIDS



Laboratory Manual

# CERTIFICATE

*This is to certify that*

*Ms. **Gopi Kota** with Enrollment No. **2303031247005** has successfully completed his laboratory experiments in the **Image Processing Lab (303105308)** from the department of **AI & AIDS** during the academic year **2025-2026**.*



**Date of Submission .....**

**Staff In charge .....**

**Head of Department.....**

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|       |                                                                                                                                                                                                                                          | From    | To |                        |                       |       |      |
| 1     | To study and implement basic commands of MATLAB required for digital image processing techniques, and various image file formats                                                                                                         |         |    |                        |                       |       |      |
| 2     | To study the effect of down sampling and quantization techniques on the grayscale image.                                                                                                                                                 |         |    |                        |                       |       |      |
| 3     | Write MATLAB code to enhance the visual quality of the image using point processing techniques, namely<br><br>a) Image negative<br><br>b) Logarithmic transformation<br><br>c) Power law transformation and gray level slicing technique |         |    |                        |                       |       |      |
| 4     | Write MATLAB code to display the histogram of the image and enhance the visual quality of the image with the help of histogram equalization technique                                                                                    |         |    |                        |                       |       |      |
| 5     | Write MATLAB code to perform the bit-plane slicing method on grayscale images.                                                                                                                                                           |         |    |                        |                       |       |      |
| 6     | Write MATLAB code to demonstrate that the convolution in the spatial domain is equivalent to multiplication in the frequency domain.                                                                                                     |         |    |                        |                       |       |      |
| 7     | Write MATLAB code to restore grayscale images from noisy images with the help of image restoration techniques.                                                                                                                           |         |    |                        |                       |       |      |

|    |                                                                                                                 |  |  |  |  |  |  |
|----|-----------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|
| 8  | Write MATLAB code to perform Min, Median & Max Filtering on grayscale image                                     |  |  |  |  |  |  |
| 9  | Write MATLAB code to blur the image using Ideal low pass, Butterworth low pass and Gaussian Low pass filter.    |  |  |  |  |  |  |
| 10 | Write MATLAB code to blur the image using Ideal High pass, Butterworth High pass and Gaussian High pass filter. |  |  |  |  |  |  |

## PRACTICAL: 01

**Aim:** To study and implement basic commands of MATLAB required for digital image processing techniques, and various image file formats.

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**Objective:** To study and implement basic MATLAB commands used in digital image processing, including reading, displaying, transforming, and analyzing images, and to understand different image file formats supported by MATLAB.

### Procedure:

#### 1. `clc`

##### Syntax:

`clc`

##### Description:

Clears the Command Window (output screen), but does not affect variables or figures.

##### Example:

`clc`

#### 2. `clear all`

##### Syntax:

`clear all`

##### Description:

Removes all variables from the workspace, freeing up memory.

##### Example:

`clear all`

### 3. close all

**Syntax:**

```
close all
```

**Description:**

Closes all open figure windows.

**Example:**

```
close all
```

### 4. imread()

**Syntax:**

```
img = imread('filename.ext');
```

**Description:**

Reads an image file into MATLAB as a matrix.

**Example:**

```
img = imread('myimage.png');
```

### 5.imwrite()

**Syntax:**

```
imwrite(img, 'filename.jpg');
```

**Description:**

Saves the image to a file.

**Example:**

```
imwrite(img, 'output_gray.jpg');
```

### 6. imshow()

**Syntax:**

```
imshow(img);
```

**Description:**

Displays an image in a figure window.

**Example:**

```
imshow(img);
```

**Output:**



**7. title()**

**Syntax:**

```
title('Image Title')
```

**Description:**

Adds a title above the current image or plot.

**Example:**

```
imshow(img);  
title('Original Image');
```

**Output:**



## 8. subplot()

**Syntax:**

```
subplot(m, n, p)
```

**Description:**

Divides the figure into an  $m \times n$  grid and activates the  $p$ th section for plotting.

**Example:**

```
subplot(1,2,1); imshow(img);  
subplot(1,2,2); imshow(rgb2gray(img));
```

**Output:**



## 9. figure

**Syntax:**

```
figure
```

**Description:**

Opens a new figure window for displaying plots/images.

**Example:**

```
figure; imshow(img);
```



**Output:**



## 10.imhist()

**Syntax:**

```
imhist(img);
```

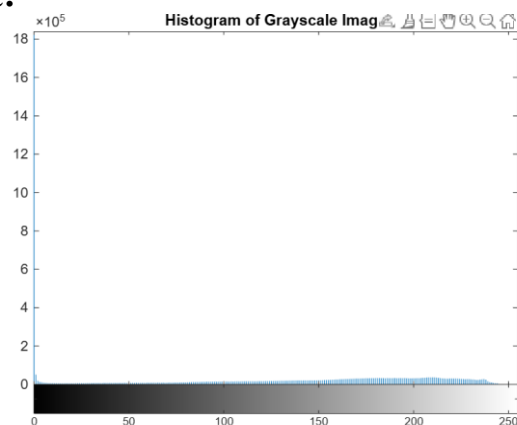
**Description:**

Displays the histogram of a grayscale image.

**Example:**

```
imhist(img);
```

**Output:**



## 11. addpath()

### Syntax:

```
addpath('folder_path')
```

### Description:

Adds a directory to MATLAB's search path to access files from that folder.

### Example:

```
addpath('C:\Users\Gopi\Documents\Images');
```

## 12. imfinfo()

### Syntax:

```
info = imfinfo(myimage.jpg')
```

### Description:

Returns metadata about the image file, such as resolution, color type, and size.

### Example:

```
addpath('C:\Users\Gopi\Documents\Images');
```

### Output:

```
How to Implement: C++ Resources for Learning C++  
Filename: '/MATLAB Drive/myimage.jpg'  
FileModDate: '24-Jun-2025 12:19:56'  
FileSize: 629635  
Format: 'jpg'  
FormatVersion: ''  
Width: 4928  
Height: 3264  
BitDepth: 24  
ColorType: 'truecolor'  
FormatSignature: ''  
NumberOfSamples: 3  
CodingMethod: 'Huffman'  
CodingProcess: 'Progressive'  
Comment: {}  
AutoOrientedWidth: 4928  
AutoOrientedHeight: 3264
```

### 13. imresize()

**Syntax:**

```
resizedImg = imresize(img, scale);
```

**Description:**

Resizes the image by a scale factor.

**Example:**

```
halfImg = imresize(img, 0.5);    % 50% of original size  
imshow(halfImg);
```

**Output:**



### 14. rgb2gray()

**Syntax:**

```
grayImg = rgb2gray(img);
```

**Description:**

Converts a color image to grayscale.

**Example:**

```
grayImg = rgb2gray(img);  
imshow(grayImg);
```

**Output:**



### 15.imadjust()

**Syntax:**

```
newImg = imadjust(img);
```

**Description:**

Enhances contrast of the image.

**Example:**

```
gray_img = rgb2gray(img);           % Convert to grayscale  
enhanced = imadjust(gray_img);  
imshow(enhanced);
```

**Output:**



## 16.edge()

### Syntax:

```
edges = edge(grayImg, 'method');
```

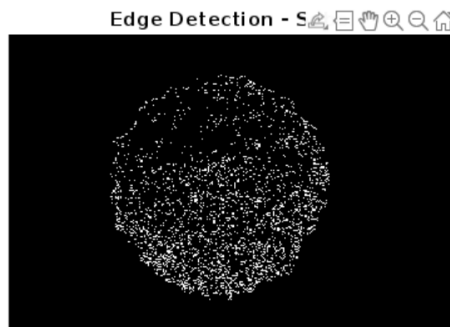
### Description:

Detects edges in the image using a specific method ( 'sobel', 'canny', etc.).

### Example:

```
gray_img = rgb2gray(img);           % Convert to grayscale  
edges = edge(gray_img, 'sobel');    % Apply Sobel edge detection  
imshow(edges);
```

### Output:



## 17.imfilter()

### Syntax:

```
filteredImg = imfilter(img, h);
```

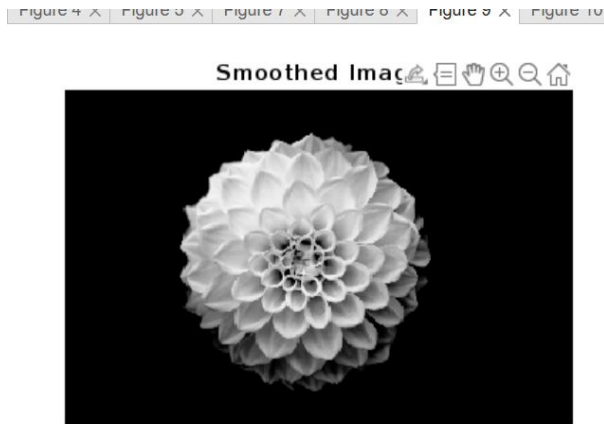
### Description:

Applies a filter to an image.

### Example:

```
h = fspecial('average', [5 5]);  
smoothImg = imfilter(gray_img, h);  
imshow(smoothImg);
```

## Output:



## 18.imnoise()

### Syntax:

```
noisyImg = imnoise(img, 'type');
```

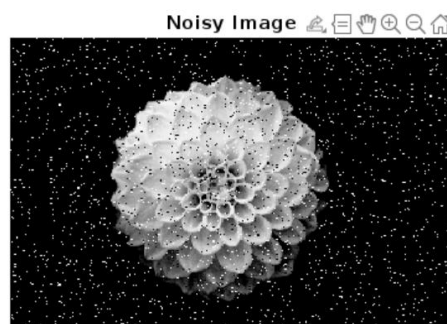
### Description:

Adds noise to the image ('salt & pepper', 'gaussian', etc.).

### Example:

```
noisy = imnoise(gray_img, 'salt & pepper');  
imshow(noisy);
```

## Output:



## 19.medfilt2()

### Syntax:

```
filtered = medfilt2(grayImg);
```

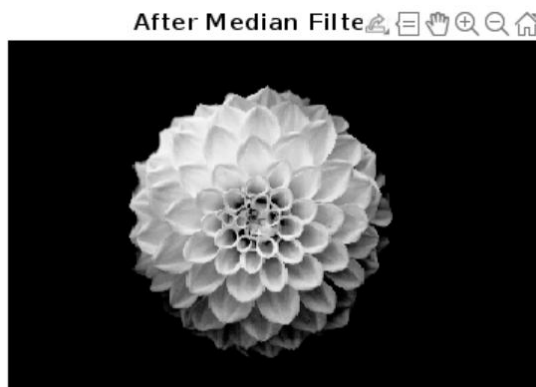
### Description:

Applies median filtering to remove noise.

### Example:

```
noisy = rgb2gray(img);  
filtered = medfilt2(noisy);  
imshow(filtered);
```

### Output:



## 20.imbinarize()

### Syntax:

```
bw = imbinarize(grayImg);
```

### Description:

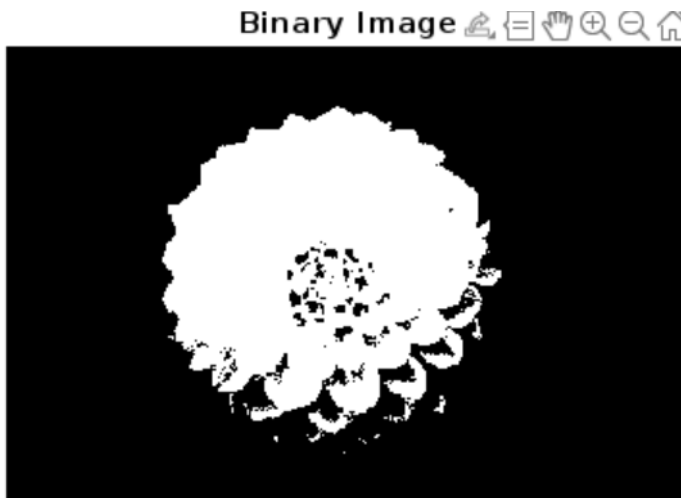
Converts a grayscale image to binary (black and white).

### Example:

```
ngrayImg = rgb2gray(img);
```

```
bw = imbinarize(grayImg);  
imshow(bw);
```

**Output:**



**21.size()**

**Syntax:**

```
[rows, cols] = size(img);
```

**Description:**

Returns the number of rows and columns in the image.

**Example:**

```
[r, c, d] = size(img);  
fprintf('Rows: %d, Columns: %d, Channels: %d\n', r, c, d);
```

**Output:**

```
AutoOrientedHeight: 3264  
  
Rows: 3264, Columns: 4928, Channels: 3  
>>
```



## Commands in brief

| Command   | Syntax             | Description                          | Example                      |
|-----------|--------------------|--------------------------------------|------------------------------|
| clc       | clc                | Clears command window                | clc                          |
| clear all | clear all          | Removes all variables from workspace | clear all                    |
| close all | close all          | Closes all figure windows            | close all                    |
| imread    | imread('file.jpg') | Reads an image from file             | img = imread('peppers.png'); |
| title     | title('text')      | Adds title to current plot           | title('Original Image');     |
| subplot   | subplot(m, n, p)   | Creates subplots in figure           | subplot(2,2,1);              |
| figure    | figure             | Opens new figure window              | figure; imshow(img);         |
| imshow    | imshow(img)        | Displays an image                    | imshow(img);                 |

| Command | Syntax                   | Description                         | Example                        |
|---------|--------------------------|-------------------------------------|--------------------------------|
| double  | double(var)              | Converts data to double precision   | A = double(img);               |
| imagesc | imagesc(data)            | Scales and displays matrix as image | imagesc(grayImg);              |
| addpath | addpath('folder')        | Adds folder to MATLAB path          | addpath('C:\images')           |
| clf     | clf                      | Clears current figure window        | clf                            |
| imwrite | imwrite(img, 'file.jpg') | Saves image to file                 | imwrite(img, 'output.jpg');    |
| imfinfo | imfinfo('file.jpg')      | Displays image file information     | info = imfinfo('peppers.png'); |
| who     | who                      | Lists variables in workspace        | who                            |
| whos    | whos                     | Lists variables with details        | whos                           |

### Conclusion:

- Basic MATLAB commands for image processing were successfully implemented. We learned how to read, display, convert, resize, and save images, and understood different image formats. This forms the foundation for further image processing techniques.